Short summary of estimation revisions

Changes on codex/rulof_new_work relative to efficient

04 September 2026

1. Baseline transition model: Table 1 and Appendix Table A1

- Re-estimated the three-wave model using the exact eight-history likelihood.
- Repaired stationarity; used 12 reproducible starts (including nested-model starts); and required optimizer convergence, a small score, positive observed information, and valid likelihood nesting before saving results.
- Replaced stale inference with analytical survey-weighted sandwich and delta standard errors.
- Separated the free initial employment probability from the model-implied long-run employment share.
- Restricted the main table to free-initial-condition specifications and moved the stationary counterparts, implied rates, and stationarity tests to A1.
- Added normalized-weight LR tests of non-zero misclassification, asymmetric error, and non-stationarity, using the boundary reference where required.

Table 1: Baseline latent-state model

	(1) No miscl.	(2) Symmetric	(3) Asymmetric
Entry rate (%)	8.28	2.98	2.98
	(0.05)	(0.06)	(0.06)
Exit rate (%)	8.54	2.91	2.91
	(0.05)	(0.06)	(0.06)
False-positive rate (%)	0.00	2.99	3.05
		(0.03)	(0.04)
False-negative rate (%)	0.00	2.99	2.92
		(0.03)	(0.04)
Employment rate (%)	49.23	50.57	50.58
	(0.18)	(0.53)	(0.53)
Likelihood-ratio tests			
LR: no misclassification	—	22,438.36	—
Boundary p-value	—	< 0.001	—
LR: symmetric errors	—	—	7.49
p-value	—	—	0.006
Log-likelihood	-357.204M	-349.371M	-349.368M
N	402,923	402,923	402,923

Note: All specifications estimate the initial employment probability freely. Rates are percentages. Symmetric error imposes equal false-positive and false-negative probabilities; the asymmetric model reports them separately. Analytical SE in parentheses use an individual-level survey-weighted sandwich covariance matrix and the delta method. Employment is the model-implied steady-state share. LR statistics use weights normalized to sum to N. The test of no misclassification uses the 50:50 boundary mixture; the asymmetry test uses χ^2_1. The calculation does not incorporate strata or PSU clustering.

Appendix Table A1: Baseline latent-state model: all specifications

	(1)	(2)	(3)	(4)	(5)	(6)
	No miscl.		Symmetric		Asymmetric	
	Stat.	Free	Stat.	Free	Stat.	Free
Transition parameters						
θ_0 (entry)	0.0818	0.0828	0.0286	0.0298	0.0286	0.0298
	(0.0004)	(0.0005)	(0.0005)	(0.0006)	(0.0005)	(0.0006)
θ_1 (persistence)	0.9136	0.9146	0.9696	0.9709	0.9696	0.9709
	(0.0004)	(0.0005)	(0.0005)	(0.0006)	(0.0005)	(0.0006)
α (initial empl.)	0.4863	0.4852	0.4850	0.4837	0.4846	0.4832
	(0.0009)	(0.0009)	(0.0009)	(0.0010)	(0.0009)	(0.0010)
Misclassification parameters						
π (symmetric)	—	—	0.0299	0.0299	—	—
			(0.0003)	(0.0003)		
π_0 (false positive)	—	—	—	—	0.0305	0.0305
					(0.0004)	(0.0004)
π_1 (false negative)	—	—	—	—	0.0293	0.0292
					(0.0004)	(0.0004)
Implied probabilities						
Entry rate (%)	8.18	8.28	2.86	2.98	2.86	2.98
	(0.04)	(0.05)	(0.05)	(0.06)	(0.05)	(0.06)
Exit rate (%)	8.64	8.54	3.04	2.91	3.04	2.91
	(0.04)	(0.05)	(0.05)	(0.06)	(0.05)	(0.06)
False-positive rate (%)	0.00	0.00	2.99	2.99	3.05	3.05
			(0.03)	(0.03)	(0.04)	(0.04)
False-negative rate (%)	0.00	0.00	2.99	2.99	2.93	2.92
			(0.03)	(0.03)	(0.04)	(0.04)
Employment rate (%)	48.63	49.23	48.50	50.57	48.46	50.58
	(0.09)	(0.18)	(0.09)	(0.53)	(0.09)	(0.53)
Likelihood-ratio tests of stationarity						
LR: stationarity	—	12.51	—	20.77	—	21.93
p-value	—	< 0.001	—	< 0.001	—	< 0.001
Log-likelihood	-357.208M	-357.204M	-349.378M	-349.371M	-349.376M	-349.368M
N	402,923	402,923	402,923	402,923	402,923	402,923

Note: Rates are percentages. Analytical SE in parentheses use an individual-level survey-weighted sandwich covariance matrix and the delta method. α is initial employment; in stationary columns it is implied by the transition parameters. Symmetric error imposes equal false-positive and false-negative rates. Stationarity LR statistics appear in the corresponding free-initial-condition column, use weights normalized to sum to N, and have a χ^2_1 reference. The calculation does not incorporate strata or PSU clustering.

Interpretation

- The no-error model implies quarterly entry and exit of about 8.2–8.6%.
- Allowing symmetric error lowers both rates to about 3%, with a per-wave misclassification probability of 2.99%.
- The no-error restriction is decisively rejected (LR 22,438.36, boundary $p < 0.001$). False-positive and false-negative probabilities remain very similar at 3.05% and 2.92%; equality is statistically rejected (LR 7.49, $p = 0.006$) but the substantive difference is only 0.13 percentage points.
- Appendix Table A1 rejects stationarity under no error, symmetric error, and asymmetric error (LR 12.51, 20.77, and 21.93; all $p < 0.001$). The restrictions nevertheless have only modest effects on the estimated transition rates.
- The repaired likelihood strengthens the paper's central conclusion: relatively infrequent clas-

sification error can create a large upward bias in measured labour-market turnover.

To do

- Check whether transition probabilities vary across wave pairs.
- Confirm the normalized-weight LR conclusions with a person-level bootstrap or likelihood profiles for the misclassification and transition rates.

2. Panel-matching robustness: Table 2

- Applied the same exact, non-stationary estimator, 12-start search, and convergence and curvature checks to panels A, B, and C.
- Held all sample filters fixed apart from progressively stricter matching.
- Added consistent analytical standard errors to every panel comparison.

Table 2: Sensitivity to alternative panel-matching rules

	Baseline (A)		Stricter (B)		Strictest (C)	
	No error	Symmetric error	No error	Symmetric error	No error	Symmetric error
Entry rate (%)	8.28	2.98	7.59	2.81	6.56	2.63
	(0.05)	(0.06)	(0.05)	(0.06)	(0.05)	(0.06)
Exit rate (%)	8.54	2.91	7.55	2.48	6.03	1.93
	(0.05)	(0.06)	(0.05)	(0.06)	(0.05)	(0.06)
Misclassification rate (%)	—	2.99	—	2.67	—	2.15
		(0.03)		(0.03)		(0.03)
Initial employment rate (%)	48.52	48.37	48.52	48.40	48.94	48.85
	(0.09)	(0.10)	(0.09)	(0.10)	(0.11)	(0.11)
Employment rate (%)	49.23	50.57	50.11	53.10	52.13	57.64
	(0.18)	(0.53)	(0.20)	(0.58)	(0.25)	(0.70)
Log-likelihood (millions)	-357.204	-349.371	-324.427	-317.258	-243.918	-238.714
N	402,923	402,923	377,132	377,132	300,200	300,200

Note: All models estimate the initial employment probability freely; stationarity is not imposed. Long-run employment is $\theta_0/(\theta_0+1-\theta_1)$. Parentheses contain individual-level survey-weighted sandwich/delta standard errors. Panels A, B, and C apply successively stricter matching rules and use otherwise identical sample filters.

Appendix Table A2a: Observable composition under alternative matching rules

	Panel A	Panel B	Panel C
N, Table 2 sample	402,923	377,132	300,200
N, complete-covariate sample	394,823	369,724	300,200
Age (years)	33.73	33.79	34.11
Education (years)	10.19	10.18	10.11
Female (%)	51.45	51.65	52.32
Black African (%)	79.87	80.43	79.20
Coloured (%)	9.54	9.29	9.79
Indian/Asian (%)	2.85	2.68	2.86
White (%)	7.75	7.61	8.14
Initially employed (%)	47.93	47.93	48.94

Note: Weighted means use each panel's survey weights. The complete-covariate samples require nonmissing age and education in all three waves and nonmissing baseline race and sex; the first row reports the larger Table 2 estimation samples.

Appendix Table A2b: Covariate-adjusted matching-rule sensitivity

	A: No error	A: Symmetric	B: No error	B: Symmetric	C: No error	C: Symmetric
Entry rate – own composition (%)	9.49	4.63	8.77	4.41	7.71	4.05
	(0.06)	(0.07)	(0.06)	(0.07)	(0.07)	(0.07)
Entry rate – Panel C composition (%)	9.55	4.66	8.81	4.43	7.71	4.05
	(0.06)	(0.07)	(0.06)	(0.07)	(0.07)	(0.07)
Exit rate – own composition (%)	11.23	5.92	10.16	5.36	8.35	4.51
	(0.08)	(0.09)	(0.08)	(0.09)	(0.08)	(0.09)
Exit rate – Panel C composition (%)	11.12	5.83	10.06	5.29	8.35	4.51
	(0.08)	(0.09)	(0.08)	(0.09)	(0.08)	(0.09)
Misclassification rate – own composition (%)	—	2.49	—	2.19	—	1.75
	(—)	(0.03)	(—)	(0.03)	(—)	(0.03)
Misclassification rate – Panel C composition (%)	—	2.49	—	2.19	—	1.75
	(—)	(0.03)	(—)	(0.03)	(—)	(0.03)
Initial employment rate (%)	47.93	47.82	47.93	47.86	48.94	48.94
N	394,823	394,823	369,724	369,724	300,200	300,200

Note: All models include Set 2 covariates (age, age squared, education, race, and sex), estimate initial employment freely, and do not impose stationarity. Panel C composition reports survey-weighted predictive margins evaluated on the same Panel C covariate distribution. Parentheses contain analytical sandwich/delta standard errors conditional on that target distribution.

Appendix Table A2c: Decomposition of matching-rule differences

	Total gap (p.p.)	Composition (p.p.)	Residual (p.p.)	Composition share (%)
A–C: No error: Entry	1.78	-0.06	1.84	-3.15
A–C: No error: Exit	2.88	0.11	2.78	3.71
A–C: Symmetric: Entry	0.58	-0.03	0.61	-4.74
A–C: Symmetric: Exit	1.40	0.09	1.31	6.28
B–C: No error: Entry	1.06	-0.05	1.11	-4.28
B–C: No error: Exit	1.81	0.10	1.71	5.50
B–C: Symmetric: Entry	0.36	-0.02	0.38	-5.75
B–C: Symmetric: Exit	0.85	0.08	0.77	9.29

Note: The composition component is the own-sample predictive margin minus the same panel model evaluated on Panel C covariates. The residual is the Panel C-standardized prediction minus Panel C's own-sample prediction. These are descriptive decompositions; the final share can be unstable when the total gap is small.

Appendix Table A2d: Pooled matching-group models

	(1) Baseline	(2) Transitions	(3) Error	(4) Both
Retained in C: entry (%)	4.62	3.60	4.27	3.98
	(0.07)	(0.07)	(0.07)	(0.07)
Retained in C: exit (%)	5.91	3.69	5.40	4.27
	(0.09)	(0.08)	(0.09)	(0.09)
Retained in C: misclassification (%)	2.50	2.00	1.58	1.70
	(0.03)	(0.03)	(0.03)	(0.03)
B but not C: entry (%)	4.62	7.50	4.27	5.64
	(0.07)	(0.14)	(0.07)	(0.16)
B but not C: exit (%)	5.91	10.03	5.40	7.93
	(0.09)	(0.17)	(0.09)	(0.19)
B but not C: misclassification (%)	2.50	2.00	4.18	3.39
	(0.03)	(0.03)	(0.06)	(0.08)
A but not B: entry (%)	4.62	14.29	4.27	8.41
	(0.07)	(0.27)	(0.07)	(0.44)
A but not B: exit (%)	5.91	21.31	5.40	15.12
	(0.09)	(0.34)	(0.09)	(0.51)
A but not B: misclassification (%)	2.50	2.00	8.63	5.82
	(0.03)	(0.03)	(0.13)	(0.23)
Log-likelihood (millions)	-337.457	-334.398	-334.397	-334.007
N	394,823	394,823	394,823	394,823
Group indicators in transitions	No	Yes	No	Yes
Group indicators in error rate	No	No	Yes	Yes

Note: All columns use Panel A, Set 2 controls, a symmetric error model, a free initial employment rate, and no stationarity restriction. Rates are predictive margins standardized to Panel A's covariate distribution while changing only the matching-group indicators. The error probability uses the bounded link $0.5\,\text{logit}^{-1}(z'\delta)$. Parentheses contain analytical sandwich/delta standard errors.

Interpretation

- Stricter matching lowers both observed and corrected transition rates.
- The estimated misclassification probability declines from 2.99% in panel A to 2.15% in panel C.
- Corrected entry and exit remain much lower than no-error rates under every matching rule.
- Matching quality is therefore part of the inconsistency mechanism, but the main correction is not driven solely by the baseline matching rule.

- Observable composition changes very little: from panel A to C, mean age rises by 0.38 years, mean education falls by 0.08 years, and the female share rises by 0.88 percentage points.
- Standardizing all models to panel C's age, education, race, and sex distribution does not remove the transition gradient. In the symmetric-error model it changes panel A entry from 4.63% to 4.66% and exit from 5.92% to 5.83%. Observable composition explains about −4.7% of the A–C entry gap and 6.3% of the exit gap; the corresponding B–C shares are −5.7% and 9.3%.
- In the pooled joint model, holding Set 2 covariates fixed, entry, exit, and misclassification are 3.98%, 4.27%, and 1.70% for records retained in C; 5.64%, 7.93%, and 3.39% for B-but-not-C records; and 8.41%, 15.12%, and 5.82% for A-but-not-B records. The stricter-match gradient therefore reflects both higher underlying mobility and higher classification error among excluded records, not ordinary demographic composition alone.
- The pooled estimates are descriptive: matching-group membership is selected by the linkage rules, so the coefficients should not be read as causal effects of applying a stricter algorithm.

To do

- Repeat the standardization with richer duration and employment covariates and test sensitivity to alternative predictive-margin target populations.
- Profile the pooled transition/error allocation and use longer-start checks; the data clearly reject equal groups, but either channel alone explains much of the raw gradient.
- Check whether the Table 4 and age/education-consistency results are also stable under the stricter matching rules.

3. Reported-characteristic inconsistencies: Table 3 and Appendix Table A3

- Re-estimated the inconsistency-indicator, severity, and matching-rule models after adding race and gender disagreement indicators to every error equation.
- Added mutually exclusive indicators for exactly two, three, or four reported- characteristic inconsistencies to each misclassification equation, with zero or one inconsistency as the omitted category.
- Retained unrestricted-initial-employment estimates in Table 3 and report the stationary counterparts alongside them in A3.
- Repaired the EM update, used multiple starts and likelihood checks, and added analytical standard errors.
- Interpreted age and education inconsistencies neutrally: they may arise from response error, matching error, or both.
- Added robustness allowing true transition rates to differ between records with and without an inconsistency and allowing misclassification to depend on inconsistency severity.
- Added stationary and free-initial-employment specifications in which the misclassification rate also depends on person-wave indicators for membership in matching panel B but not C and panel A but not B, defined using the Table 2 sample-construction rules.
- Reduced the main table to the inconsistency-indicator, severity plus matching-rule, and transition-covariate specifications. A3 now contains only the stationary/free pairs for these same three specifications; the severity-only and Set 2-only results remain in separate appendix outputs.

- Added a conditional-stationarity counterpart to the preferred model. It makes wave-1 initial employment depend on the wave-1 covariate-specific entry and exit hazards, while retaining the wave-varying transition equations.
- Simplified A3 by replacing pattern-specific probabilities and probability increments with the lowest, median, mean, and highest predicted misclassification probabilities, with analytical standard errors. It also reports the preferred-model transition and error coefficients under conditional stationarity and free initial employment.

Table 3: Reported-characteristic inconsistencies

	(1) Inconsistency indicators	(2) + severity and matching-rule indicators	(3) + transition covariates and inconsistency effects
Entry rate (%)	2.56**	2.52**	3.54**
	(0.05)	(0.05)	(0.05)
Exit rate (%)	2.14**	2.09**	3.51**
	(0.06)	(0.06)	(0.05)
Misclassification rate (%)	3.38**	3.42**	2.85**
	(0.03)	(0.03)	(0.03)
Initial employment rate (%)	47.60**	47.60**	47.88**
	(0.10)	(0.10)	(0.10)
Log-likelihood	-332.842M	-332.440M	-329.147M
N	394,823	394,823	394,823
Inconsistency-severity effects	No	Yes	No
Matching-rule indicators	No	Yes	No
Set 2 transition covariates	No	No	Yes
Inconsistency effects in transitions	No	No	Yes

Note: All specifications estimate initial employment freely. Each misclassification equation includes age, education, race, and gender inconsistency indicators plus mutually exclusive indicators for exactly two, three, or four inconsistencies (zero or one is omitted); race and gender disagreements are attributed to waves using the same adjacent-wave logic as matching rule B in Table 2. Column (2) adds age and education severity effects and the matching-rule indicators. Column (3) instead combines this error equation with Set 2 transition covariates and the four origin-wave inconsistency effects in both entry and persistence. Its entry and exit rates are posterior risk-set, survey-weighted means. Mean misclassification averages individual-wave predictions using survey weights. Analytical standard errors use the survey-weighted sandwich covariance and delta method. An inconsistency may arise from response error, matching error, or both.

Appendix Table A3, Panel A: Reported-characteristic inconsistencies: implied rates and error equation

	(1)	(2)	(3)	(4)	(5)	(6)
	Inconsistency indicators		+ severity and matching-rule indicators		+ transition covariates and inconsistency effects	
	Stationary	Free α	Stationary	Free α	Conditional stat.	Free α
Implied rates						
Entry rate (%)	2.27**	2.56**	2.23**	2.52**	3.47**	3.54**
	(0.05)	(0.05)	(0.04)	(0.05)	(0.05)	(0.05)
Exit rate (%)	2.47**	2.14**	2.42**	2.09**	3.78**	3.51**
	(0.05)	(0.06)	(0.05)	(0.06)	(0.05)	(0.05)
Initial employment rate (%)	47.93**	47.60**	47.92**	47.60**	47.93**	47.88**
	(0.09)	(0.10)	(0.09)	(0.10)	(0.08)	(0.10)
Employment-state transformations						
Steady-state employment (%)	47.93**	54.47**	47.92**	54.72**	—	—
	(0.09)	(0.63)	(0.09)	(0.64)		
Initial minus steady employment (p.p.)	0.00	-6.86**	0.00	-7.12**	—	—
	(fixed)	(0.65)	(fixed)	(0.66)		
Distribution of predicted misclassification probabilities						
Lowest (%)	2.22**	2.22**	2.02**	2.02**	1.74**	1.79**
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Median (%)	2.22**	2.22**	2.02**	2.02**	1.74**	1.79**
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Mean (%)	3.38**	3.38**	3.42**	3.42**	2.71**	2.85**
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Highest (%)	39.06**	39.20**	42.14**	42.23**	35.19**	40.66**
	(4.45)	(4.45)	(4.78)	(4.78)	(0.83)	(5.08)
Logistic-link coefficients						
δ_0 (intercept)	-3.0690**	-3.0686**	-3.1695**	-3.1692**	-3.3246**	-3.2962**
	(0.0150)	(0.0150)	(0.0160)	(0.0160)	(0.0161)	(0.0154)
δ_1 (age inconsistency)	2.0365**	2.0393**	0.7982**	0.7979**	2.0173**	2.0762**
	(0.0324)	(0.0323)	(0.0613)	(0.0612)	(0.0386)	(0.0378)
δ_2 (education inconsistency)	1.0820**	1.0832**	0.7434**	0.7431**	1.0069**	1.0371**
	(0.0253)	(0.0252)	(0.0384)	(0.0383)	(0.0327)	(0.0319)
δ_3 (race inconsistency)	0.0821	0.0907	-0.2519**	-0.2454**	0.0312	0.1844$^{\cdot}$
	(0.0879)	(0.0880)	(0.0940)	(0.0941)	(0.1077)	(0.1105)
δ_4 (gender inconsistency)	1.4929**	1.4949**	1.1774**	1.1776**	1.3495**	1.4595**
	(0.0424)	(0.0424)	(0.0539)	(0.0538)	(0.0514)	(0.0503)
Exactly two inconsistencies	0.2542**	0.2530**	0.4329**	0.4317**	0.3653**	0.4143**
	(0.0512)	(0.0511)	(0.0565)	(0.0564)	(0.0601)	(0.0601)
Exactly three inconsistencies	-0.3420**	-0.3429**	0.0664	0.0657	-0.1832$^{\cdot}$	-0.0132
	(0.1010)	(0.1010)	(0.1101)	(0.1102)	(0.1099)	(0.1209)
Exactly four inconsistencies	-0.3513	-0.3509	0.6219	0.6215	-0.3650	0.0103
	(0.5332)	(0.5385)	(0.7317)	(0.7384)	(0.4996)	(0.6817)
δ_5 (large age increment)	0.00	0.00	1.2585**	1.2619**	0.00	0.00
	(fixed)	(fixed)	(0.0619)	(0.0618)	(fixed)	(fixed)
δ_6 (large education increment)	0.00	0.00	0.0838*	0.0861*	0.00	0.00
	(fixed)	(fixed)	(0.0418)	(0.0417)	(fixed)	(fixed)
δ_7 (panel B but not C)	0.000	0.000	0.4315**	0.4316**	—	—
	(fixed)	(fixed)	(0.0220)	(0.0220)		
δ_8 (panel A but not B)	0.000	0.000	0.4170**	0.4191**	—	—
	(fixed)	(fixed)	(0.0338)	(0.0336)		
Log-likelihood	-332.895M	-332.842M	-332.495M	-332.440M	-299.287M	-329.147M
N	394,823	394,823	394,823	394,823	394,823	394,823
Stationarity imposed	Yes	No	Yes	No	Conditional	No
Inconsistency-severity effects	No	No	Yes	Yes	No	No
Matching-rule indicators	No	No	Yes	Yes	No	No
Set 2 transition covariates	No	No	No	No	Yes	Yes
Inconsistency effects in transitions	No	No	No	No	Yes	Yes

Note: The table retains stationary and free-initial-employment versions of the same three specifications shown in Table 3. Misclassification summaries describe the survey-weighted distribution of predicted person-wave probabilities over the estimation sample; lowest and highest are the observed-support extrema. Error-equation coefficients are on the half-logit index. Exact-count indicators are mutually exclusive and omit zero or one inconsistency. In the conditionally stationary preferred model, each person's wave-1 initial employment probability is implied by their wave-1 covariate-specific entry and exit hazards; later hazards may change with origin-wave inconsistencies. Parentheses contain analytical sandwich/delta standard errors.

Appendix Table A3, Panel B: Transition-covariate specification: transition coefficients

	Conditional stat.	Free α
Entry coefficients (probit index)		
intercept	-1.7794**	-1.6867**
	(0.0091)	(0.0099)
age	1.7902**	1.9365**
	(0.0326)	(0.0466)
age sq	-1.6657**	-1.8277**
	(0.0327)	(0.0496)
educ	-0.0147**	0.0632**
	(0.0038)	(0.0070)
race 2	0.2403**	0.1653**
	(0.0129)	(0.0195)
race 3	-0.2074**	-0.3627**
	(0.0315)	(0.0624)
race 4	-0.1286**	-0.1709**
	(0.0264)	(0.0422)
female	-0.2060**	-0.3062**
	(0.0087)	(0.0123)
origin age inconsistency	0.4790**	0.3667**
	(0.0214)	(0.0340)
origin education inconsistency	0.2354**	0.1984**
	(0.0147)	(0.0215)
origin race inconsistency	0.1489*	0.1191
	(0.0589)	(0.0827)
origin gender inconsistency	0.2337**	-0.0118
	(0.0291)	(0.0596)
Persistence coefficients (probit index)		
intercept	1.8302**	1.8428**
	(0.0089)	(0.0117)
age	0.2514**	1.2098**
	(0.0362)	(0.0573)
age sq	-0.0395	-0.8621**
	(0.0355)	(0.0565)
educ	0.2300**	0.2084**
	(0.0042)	(0.0054)
race 2	-0.0177	0.1153**
	(0.0123)	(0.0205)
race 3	0.3304**	0.8784**
	(0.0294)	(0.1182)
race 4	0.4434**	0.8025**
	(0.0231)	(0.0672)
female	-0.1240**	-0.1403**
	(0.0087)	(0.0135)
origin age inconsistency	-0.5502**	-0.5880**
	(0.0225)	(0.0295)
origin education inconsistency	-0.2395**	-0.3119**
	(0.0150)	(0.0212)
origin race inconsistency	-0.0211	0.2490
	(0.0554)	(0.1632)
origin gender inconsistency	-0.3670**	-0.3078**
	(0.0306)	(0.0427)

Note: The conditional-stationarity column derives each person's wave-1 initial employment probability from their wave-1 covariate-specific entry and exit hazards; the free column estimates one common initial probability. Parentheses contain analytical sandwich standard errors.

Appendix Table A3, Panel B (continued): Transition-covariate specification: misclassification coefficients and diagnostics

	Conditional stat.	Free α
Misclassification coefficients (half-logit index)		
Intercept	-3.3246**	-3.2962**
	(0.0161)	(0.0154)
Age inconsistency	2.0173**	2.0762**
	(0.0386)	(0.0378)
Education inconsistency	1.0069**	1.0371**
	(0.0327)	(0.0319)
Race inconsistency	0.0312	0.1844·
	(0.1077)	(0.1105)
Gender inconsistency	1.3495**	1.4595**
	(0.0514)	(0.0503)
Exactly two inconsistencies	0.3653**	0.4143**
	(0.0601)	(0.0601)
Exactly three inconsistencies	-0.1832·	-0.0132
	(0.1099)	(0.1209)
Exactly four inconsistencies	-0.3650	0.0103
	(0.4996)	(0.6817)
Initial state and diagnostics		
Mean initial employment probability	0.4793**	0.4788**
	(0.0008)	(0.0010)
Log-likelihood	-299.287M	-329.147M
Information rank	32/32	33/33
Information condition number	281717	33346
N	394,823	394,823

Note: The conditional-stationarity column derives each person's wave-1 initial employment probability from their wave-1 covariate-specific entry and exit hazards; the free column estimates one common initial probability. The likelihoods are not nested because only the former allows observed initial-state heterogeneity. Parentheses contain analytical sandwich/delta standard errors.

Appendix Table A3A: Transition-covariate specifications

	(A) Set 2 transition covariates	(B) + inconsistency effects, free α
Origin-wave inconsistency coefficients: entry		
Age inconsistency	—	0.3667^{**}
		(0.0340)
Education inconsistency	—	0.1984^{**}
		(0.0215)
Race inconsistency	—	0.1191
		(0.0827)
Gender inconsistency	—	-0.0118
		(0.0596)
Origin-wave inconsistency coefficients: persistence		
Age inconsistency	—	-0.5880^{**}
		(0.0295)
Education inconsistency	—	-0.3119^{**}
		(0.0212)
Race inconsistency	—	0.2490
		(0.1632)
Gender inconsistency	—	-0.3078^{**}
		(0.0427)
Misclassification-equation coefficients		
Intercept	-3.3098^{**}	-3.2962^{**}
	(0.0158)	(0.0154)
Age inconsistency	2.2180^{**}	2.0762^{**}
	(0.0344)	(0.0378)
Education inconsistency	1.1907^{**}	1.0371^{**}
	(0.0284)	(0.0319)
Race inconsistency	0.2907^{**}	$0.1844^{\cdot}$
	(0.0951)	(0.1105)
Gender inconsistency	1.6290^{**}	1.4595^{**}
	(0.0452)	(0.0503)
Implied misclassification probabilities		
No inconsistency (%)	1.76^{**}	1.79^{**}
	(0.03)	(0.03)
Age inconsistency (%)	12.56^{**}	11.40^{**}
	(0.29)	(0.31)
Education inconsistency (%)	5.36^{**}	4.73^{**}
	(0.12)	(0.13)
Race inconsistency (%)	2.33^{**}	2.13^{**}
	(0.21)	(0.22)
Gender inconsistency (%)	7.85^{**}	6.87^{**}
	(0.29)	(0.29)
Log-likelihood	-329.474M	-329.147M
Information rank	25/25	33/33
Information condition number	28492	33346
N	394,823	394,823

Note: Transition equations use probit links; the misclassification equation uses the half-logit link. Column (B) adds origin-wave inconsistency effects to both entry and persistence. The complete coefficients for this specification under free initial employment and conditional stationarity appear in Appendix Table A3, Panel B. Analytical standard errors use the survey-weighted sandwich covariance and delta method.

The additional reliability-group robustness table is generated as table_inconsistency_reliability.tex in EM-baseline-ext/output/tables.

Interpretation

- In the unrestricted indicators-only model, mean misclassification is 3.39%; the entry and exit rates are 2.57% and 2.15%, and initial employment is 47.60%.
- Conditional on the other indicators, a race disagreement barely changes predicted error in the first two models. A gender disagreement is much more informative: predicted error rises from 2.21% to 8.85% in the indicators-only model.
- Stationarity materially changes the entry and exit decomposition but barely changes the inconsistency-related misclassification probabilities. In the free model, initial employment is 6.56 percentage points below its implied steady-state rate.
- Allowing additional large-increment effects substantially sharpens the age gradient: predicted misclassification rises from 6.31% for a mild age inconsistency to 16.66% for a large one. The corresponding education increase is only 0.62 percentage points.
- In the augmented free-initial-employment model, predicted misclassification for a record with no reported-characteristic inconsistency is 2.01% when retained by both stricter panels and 2.93% in either B-but-not-C or A-but-not-B observations. Thus the matching-rule indicators retain a reliability gradient after controlling for race and gender disagreements.
- Adding the matching-rule indicators attenuates but does not eliminate the age/education gradients: predicted error is 4.79% for a mild and 13.89% for a large age inconsistency, and 4.34% and 4.77% for mild and large education inconsistencies. Mean error changes little, from 3.40% to 3.41%; entry and exit move from 2.57% and 2.14% to 2.54% and 2.10%.
- In the matching-rule-indicator model, a race disagreement predicts 1.70% error and a gender disagreement 6.67%, compared with 2.01% when none of the four reported characteristics is inconsistent. The race contrast is small and negative; the gender contrast remains large and positive.
- The omitted Set 2-only main-table column remains in A3A: its posterior risk-set mean entry and exit rates are 3.37% and 3.17%, while mean misclassification is 2.93%.
- In the preferred free-initial model, adding the inconsistency indicators to both transition equations raises the mean entry and exit rates to 3.55% and 3.51% and lowers mean error only modestly, to 2.85%. The likelihood gain is large, so the indicators capture real transition heterogeneity as well as measurement reliability.
- Under conditional stationarity, the corresponding entry, exit, and mean misclassification rates are 3.47% (SE 0.05), 3.78% (0.05), and 2.71% (0.03); mean initial employment is 47.93% (0.08). This model is full rank (29/29). Its much larger likelihood is not an LR test of stationarity: unlike the free model's common initial probability, conditional stationarity lets initial employment vary with observed covariates.
- The error gradient remains economically large after this control: predicted error is 1.76% with no inconsistency, 12.52% with an age inconsistency, 5.01% with an education inconsistency, and 7.53% with a gender inconsistency.
- The relationship is not explained solely by different true transition rates among records with age or education inconsistencies.
- Larger age inconsistencies are much more informative than mild ones; inconsistency severity matters less for education.
- Age, education, and gender inconsistencies predict higher status misclassification, without distinguishing response errors from incorrect panel matches.

To do

- Check alternative nonlinear definitions of inconsistency extent and more flexible inconsistency-group transition rates.
- Bootstrap the age/education-inconsistency effects and repeat the analysis under stricter panel matching.
- Test alternative timing conventions for putting inconsistency indicators in the transition equations and allow the initial employment probability to depend on baseline covariates.

4. Observable transition heterogeneity: Table 4

- Repaired the covariate-model optimiser so every specification maximises the appropriate likelihood, uses five starts including a nested-model warm start, and dominates its nested no-error model.
- Added analytical standard errors and posterior risk-set and survey-weighted transition rates.
- Retained the old Set 2 as new Set 1 and the old Set 4 as new Set 2; omitted the old Sets 1 and 3 to focus the table on the two preferred nested sets.
- New Set 2 adds tenure, time since work, time-varying contract type, and informal-sector status without restricting the sample to respondents observed in employment at every wave.
- Added a new Set 2 specification in which the symmetric misclassification rate uses exactly the Table 3 column (1) covariates: age, education, race, and gender inconsistency indicators plus indicators for exactly two, three, or four inconsistencies.
- Moved raw coefficients, the initial-state parameter, total churn, and hazard distributions to an appendix table.

Table 4: Risk-set and survey-weighted quarterly employment transitions

	(1)	(2)	(3)	(4)	(5)
	No miscl.		Constant symmetric		Reliability-dependent
	Set 1	Set 2	Set 1	Set 2	Set 2
Entry rate (%)	8.14	8.14	3.77	4.87	4.67
	(0.05)	(0.05)	(0.06)	(0.06)	(0.06)
Exit rate (%)	8.58	8.58	3.89	5.09	4.62
	(0.05)	(0.05)	(0.06)	(0.06)	(0.06)
Misclassification rate (%)	—	—	2.50	2.18	2.55
			(0.03)	(0.03)	(0.03)
Employment rate (%)	47.94	47.94	47.84	47.87	47.88
	(0.09)	(0.09)	(0.09)	(0.09)	(0.01)
Log-likelihood	-342.6M	-336.2M	-337.5M	-332.7M	-324.0M
N	394,823	394,823	394,823	394,823	394,823
Covariates included					
Age, Age^2, Educ.	Yes	Yes	Yes	Yes	Yes
+ Race, Gender	Yes	Yes	Yes	Yes	Yes
+ Tenure (exit), time since work (entry)	No	Yes	No	Yes	Yes
+ Contract type, informal sector (exit only)	No	Yes	No	Yes	Yes
Reliability variables in error equation	No	No	No	No	Yes

Note: All specifications estimate the initial employment probability freely. Rates average predicted hazards using survey weights and posterior probabilities of belonging to the relevant origin-state risk set. Employment is the survey-weighted posterior employed share. SE are bootstrap estimates when available and otherwise use an individual-level survey-weighted sandwich/delta approximation. The approximation does not incorporate strata or PSU clustering. The retained old Sets 2 and 4 are relabelled Sets 1 and 2. Column (5) retains the new Set 2 transition equations and uses the same symmetric-error covariates as Table 3, column (1): age, education, race, and gender inconsistency indicators plus indicators for exactly two, three, or four inconsistencies.

Appendix Table A4: Covariate-model coefficients and predicted-hazard distributions

	(1)	(2)	(3)	(4)
	No miscl.		Symmetric	
	Set 1	Set 2	Set 1	Set 2
Other derived model quantities				
Initial employment rate (%)	47.93**	47.93**	47.82**	47.87**
	(0.09)	(0.09)	(0.10)	(0.10)
Total employment churn (%)	8.35**	8.35**	3.83**	4.98**
	(0.04)	(0.04)	(0.05)	(0.05)
Entry equation: $\Phi(X_{it}\beta_0)$				
Intercept	-1.2464**	-0.8707**	-1.6189**	-1.0416**
	(0.0053)	(0.0069)	(0.0093)	(0.0097)
Age	1.4214**	1.3364**	1.9325**	1.4300**
	(0.0241)	(0.0266)	(0.0454)	(0.0490)
Age^2	-1.3104**	-1.2092**	-1.8232**	-1.3505**
	(0.0249)	(0.0264)	(0.0481)	(0.0480)
Education	0.0386**	0.0071·	0.0483**	-0.0137*
	(0.0038)	(0.0039)	(0.0068)	(0.0063)
Race: Coloured	0.1472**	0.0631**	0.1508**	0.0715**
	(0.0111)	(0.0113)	(0.0190)	(0.0169)
Race: Indian	-0.1557**	-0.1096**	-0.2941**	-0.2390**
	(0.0263)	(0.0269)	(0.0571)	(0.0514)
Race: White	-0.0678**	-0.0532*	-0.1598**	-0.1506**
	(0.0207)	(0.0211)	(0.0409)	(0.0368)
Female	-0.2247**	-0.1599**	-0.2948**	-0.2015**
	(0.0068)	(0.0069)	(0.0120)	(0.0111)
log(1 + tenure)	—	0.0000	—	0.0000
		[fixed]		[fixed]
log(1 + timesincework)	—	-0.2455**	—	-0.3071**
		(0.0043)		(0.0064)
Never worked	—	-0.5413**	—	-0.8352**
		(0.0086)		(0.0212)
Tenure missing	—	0.0000	—	0.0000
		[fixed]		[fixed]
Permanent contract	—	0.0000	—	0.0000
		[fixed]		[fixed]
Informal sector	—	0.0000	—	0.0000
		[fixed]		[fixed]
Employment-persistence equation: $\Phi(X_{it}\beta_1)$				
Intercept	1.2772**	1.0437**	1.6890**	1.2854**
	(0.0053)	(0.0085)	(0.0103)	(0.0130)
Age	0.9015**	0.6915**	1.0118**	0.6434**
	(0.0309)	(0.0318)	(0.0543)	(0.0505)
Age^2	-0.6602**	-0.5799**	-0.6765**	-0.5117**
	(0.0298)	(0.0307)	(0.0536)	(0.0499)
Education	0.1756**	0.1049**	0.2118**	0.1230**
	(0.0034)	(0.0038)	(0.0051)	(0.0058)
Race: Coloured	0.0969**	0.0197·	0.1348**	-0.0168
	(0.0104)	(0.0107)	(0.0191)	(0.0168)
Race: Indian	0.3125**	0.2307**	0.6983**	0.5341**
	(0.0250)	(0.0262)	(0.0861)	(0.0626)
Race: White	0.3926**	0.3083**	0.8114**	0.6342**
	(0.0168)	(0.0175)	(0.0669)	(0.0470)
Female	-0.0973**	-0.0845**	-0.1181**	-0.0678**
	(0.0070)	(0.0072)	(0.0123)	(0.0115)
log(1 + tenure)	—	0.2998**	—	0.5083**
		(0.0049)		(0.0120)
log(1 + timesincework)	—	0.0000	—	0.0000
		[fixed]		[fixed]
Never worked	—	0.0000	—	0.0000
		[fixed]		[fixed]
Tenure missing	—	-0.0398	—	-0.0993
		(1.0311)		(1.2935)
Permanent contract	—	0.2191**	—	0.2657**
		(0.0086)		(0.0135)
Informal sector	—	-0.1943**	—	-0.2417**
		(0.0094)		(0.0136)
Distribution of predicted entry hazards				
10th percentile	3.67	3.20	0.97	1.00
Median	7.86	5.99	3.29	2.21
90th percentile	13.59	17.44	7.62	13.82
Distribution of predicted exit hazards				
10th percentile	3.29	1.94	0.41	0.19
Median	7.69	6.78	3.00	2.51
90th percentile	14.84	18.05	8.40	14.29
Set 2 discrete effects on the exit probability				
Permanent contract	—	-3.28**	—	-2.43**
		(0.13)		(0.13)
Informal sector	—	3.01**	—	2.36**
		(0.16)		(0.15)

Note: Raw probit coefficients are reported. Tenure enters persistence only; time since work and never worked enter entry only. Contract type and sector are origin-wave characteristics and may change between transitions; their entry coefficients are fixed at zero because these attributes are not defined for non-employed origin states. Hazard percentiles use survey and posterior risk-set weights. SE are bootstrap estimates when available and otherwise use an individual-level survey-weighted sandwich/delta approximation; quantile SE are omitted.

Appendix Table A4B: Reliability-dependent misclassification estimates for Table 4

	Set 2; reliability-dependent error
Symmetric-error equation: $0.5\,\Lambda(Z_{it}\delta)$	
Intercept	-3.6366
	(0.0235)
Age inconsistency	2.6257
	(0.0393)
Education inconsistency	1.4840
	(0.0345)
Race inconsistency	0.4872
	(0.1091)
Gender inconsistency	1.9152
	(0.0519)
Exactly two inconsistencies	0.0217
	(0.0614)
Exactly three inconsistencies	-0.7792
	(0.1351)
Exactly four inconsistencies	-0.9243
	(0.9093)
Distribution of predicted misclassification probabilities	
Lowest (%)	1.28
	(0.03)
Median (%)	1.28
	(0.03)
Mean (%)	2.55
	(0.03)
Highest (%)	43.78
	(4.89)

Note: Parentheses contain analytical individual-level survey-weighted sandwich/delta-method standard errors. The error equation matches Table 3, column (1). Exact-count indicators are mutually exclusive, with zero or one inconsistency as the omitted category. Distribution summaries use survey weights over all person-wave observations.

Interpretation

- No-error entry and exit remain close to 8.1% and 8.6% across covariate sets.
- Symmetric-error estimates range from 3.77–4.87% for entry and 3.89–5.09% for exit; estimated misclassification falls from 2.50% in Set 1 to 2.18% in Set 2.
- In symmetric Set 2, predicted risks are highly dispersed: the weighted 10th–90th percentile range is 1.00–13.82% for entry and 0.19–14.29% for exit.
- Allowing error rates to depend on linked-record reliability gives entry and exit rates of 4.67% and 4.62%, and raises the mean error rate from 2.18% in constant-error Set 2 to 2.55%.
- Predicted error has a weighted median of 1.28% and mean of 2.55%, but reaches 43.78% at the observed maximum. Within inconsistency-count groups, the survey-weighted means rise from 1.28% with no inconsistency to 6.79%, 30.68%, 40.78%, and 43.78% with one through four inconsistencies. Reliability heterogeneity is therefore substantial, although corrected entry and exit remain close to the constant-error Set 2 values.
- Observable heterogeneity therefore explains some differences in transition risk but does not explain away the classification-error signal.
- Duration is economically important: entry falls with time since work and exit falls with tenure.
- Permanent contracts reduce exit and informal employment raises it. These are conditional associations, not causal estimates.

- The paper should distinguish heterogeneous true transition risk from the upward bias in observed transitions caused by classification error.

To do

- Allow the initial employment probability to depend on baseline covariates and check whether this changes the transition and error estimates.
- Test more flexible duration functions and confirm the main inference with a person-level bootstrap.
- Check robustness to alternative definitions and interactions for the reliability indicators in the misclassification equation.

5. Higher-order state dependence: Table 5

- Added the comparable four-wave AR(1) benchmark with symmetric error and free initial employment.
- Added the same tenure and time-since-work controls used in Table 4.
- Allowed duration covariates to change across origin waves.
- Added a nested AR(2) extension with the full Table 4 Set 2 transition controls, followed by a single model whose symmetric-error predictors match Table 3 column (1).
- Replaced the log-linear duration terms with piecewise-constant hazards using 0–3, 4–6, 7–12, 13–24, 25–60, and over-60-month categories.
- Used informed and perturbed starts, exact stage-specific likelihood collapse, likelihood-nesting checks, Newton polishing, and score/rank diagnostics.
- Replaced approximate optimisation and inference with exact collapsed likelihood estimation and analytical standard errors.

Table 5: Four-wave AR(1) and AR(2) estimates

	AR(1)	AR(2)	AR(2), duration controls	AR(2), Set 2	AR(2), Set 2, duration bins	+ Table 3 column (1) error predictors
Entry rate: nonemployed in both preceding waves	3.20	4.60	4.51	4.72	4.73	4.32
	(0.04)	(0.06)	(0.07)	(0.06)	(0.06)	(0.05)
Entry rate: employed two waves ago	3.20	30.75	17.48	31.13	31.34	28.98
	(0.04)	(0.47)	(0.97)	(0.53)	(0.53)	(0.72)
Exit rate: nonemployed two waves ago	3.13	31.60	19.16	32.37	32.25	30.18
	(0.04)	(0.48)	(0.81)	(0.50)	(0.51)	(0.64)
Exit rate: employed in both preceding waves	3.13	4.38	4.27	4.56	4.59	4.03
	(0.04)	(0.06)	(0.08)	(0.06)	(0.06)	(0.05)
Misclassification rate	2.70	0.98	1.49	0.93	0.90	1.79
	(0.03)	(0.04)	(0.08)	(0.05)	(0.05)	(0.06)
Initial employment rate	47.97	48.11	48.10	47.83	47.83	47.94
	(0.11)	(0.10)	(0.11)	(0.11)	(0.11)	(0.11)
Log likelihood	-468,443.8	-467,392.7	-462,205.1	-445,472.5	-445,316.0	-433,958.4
N	317,437	317,437	317,437	308,240	308,240	308,240
Parameters	4	8	13	29	37	36
Second employment lag	No	Yes	Yes	Yes	Yes	Yes
Duration controls	No	No	Yes	Yes	Yes	Yes
Duration form	—	—	Log-linear	Log-linear	Piecewise constant	Log-linear
Set 2 transition controls	No	No	No	Yes	Yes	Yes
Misclassification predictors	Constant	Constant	Constant	Constant	Constant	Table 3 column (1)

Notes. Entries are model-implied quarterly probabilities in percent. Covariate specifications report posterior risk-set and survey-weighted averages; the misclassification row is the corresponding posterior mean probability. Initial employment is the latent employment probability in wave 2. Parentheses contain analytical sandwich/delta-method standard errors. The AR(1) rate is repeated across the two corresponding AR(2) rows because the first-order model imposes equality across the second-lag state. Every specification has one latent type, symmetric misclassification, and free initial conditions. Set 2 includes the Table 4 demographic, tenure, time-since-work, contract, and informal-sector transition controls. Piecewise duration uses 0–3 months as the omitted category, followed by 4–6, 7–12, 13–24, 25–60, and over 60 months; never-worked and missing durations remain separate. The final error equation uses age, education, race, and gender inconsistency indicators plus mutually exclusive indicators for exactly two, three, or four inconsistencies, matching Table 3 column (1). The three Set 2 columns use the same 308,240 observations. The log-linear constant-error and final error models are nested; the piecewise and log-linear duration forms are non-nested.

Appendix Table A5: Four-wave AR(1) and AR(2) coefficients

Coefficient	AR(1)	AR(2)	AR(2), duration controls
Initial employment logit	-0.081	—	—
	(0.004)		
Initial-pair logit: 10	—	-2.677	-2.847
		(0.016)	(0.031)
Initial-pair logit: 01	—	-2.674	-2.839
		(0.016)	(0.030)
Initial-pair logit: 11	—	-0.081	-0.081
		(0.004)	(0.004)
Entry: intercept	-1.852	-1.685	-1.207
	(0.006)	(0.006)	(0.012)
Entry: employed two waves ago	—	1.182	0.811
		(0.014)	(0.034)
Entry: log time since work	—	—	-0.215
			(0.010)
Entry: never worked	—	—	-0.831
			(0.028)
Entry: timegap missing	—	—	-0.132
			(0.082)
Persistence: intercept	1.862	0.479	0.575
	(0.006)	(0.013)	(0.025)
Persistence: employed two waves ago	—	1.230	0.903
		(0.014)	(0.025)
Persistence: log tenure	—	—	0.431
			(0.015)
Persistence: tenure missing	—	—	0.003
			(0.719)
Misclassification logit	-2.862	-3.917	-3.486
	(0.010)	(0.042)	(0.055)

Appendix Table A5B: Set 2 AR(2) extension coefficients

Coefficient	Set 2, log-linear	Set 2, duration bins	+ Table 3 column (1) error predictors
Initial-pair logit: 10	-2.696	-2.686	-3.039
	(0.019)	(0.020)	(0.029)
Initial-pair logit: 01	-2.685	-2.676	-2.921
	(0.019)	(0.019)	(0.025)
Initial-pair logit: 11	-0.094	-0.094	-0.093
	(0.004)	(0.004)	(0.004)
Entry: intercept	-1.154	-1.036	-1.115
	(0.011)	(0.020)	(0.011)
Entry: employed two waves ago	0.860	0.869	0.716
	(0.019)	(0.020)	(0.028)
Entry: age	1.113	0.862	1.106
	(0.038)	(0.036)	(0.041)
Entry: age squared	-1.045	-0.821	-1.038
	(0.037)	(0.036)	(0.041)
Entry: education	0.005	0.017	0.002
	(0.005)	(0.005)	(0.006)
Entry: race 2	0.031	0.031	0.019
	(0.015)	(0.015)	(0.016)
Entry: race 3	-0.151	-0.156	-0.193
	(0.041)	(0.041)	(0.044)
Entry: race 4	-0.105	-0.103	-0.123
	(0.032)	(0.032)	(0.036)
Entry: female	-0.143	-0.147	-0.139
	(0.010)	(0.010)	(0.010)
Entry: log time since work	-0.215	—	-0.285
	(0.007)		(0.009)
Entry: never worked	-0.585	-0.746	-0.730
	(0.015)	(0.024)	(0.021)

Coefficient	Set 2, log-linear	Set 2, duration bins	+ Table 3 column (1) error predictors
Entry: timegap missing	-0.332	-0.306	-0.422
	(0.201)	(0.206)	(0.206)
Persistence: intercept	0.471	0.276	0.553
	(0.018)	(0.018)	(0.024)
Persistence: employed two waves ago	0.882	0.841	0.767
	(0.017)	(0.020)	(0.024)
Persistence: age	0.567	0.521	0.539
	(0.044)	(0.043)	(0.048)
Persistence: age squared	-0.468	-0.403	-0.465
	(0.043)	(0.042)	(0.047)
Persistence: education	0.101	0.101	0.086
	(0.005)	(0.005)	(0.006)
Persistence: race 2	-0.010	-0.003	-0.042
	(0.015)	(0.015)	(0.016)
Persistence: race 3	0.270	0.265	0.308
	(0.041)	(0.041)	(0.045)
Persistence: race 4	0.381	0.380	0.402
	(0.032)	(0.031)	(0.036)
Persistence: female	-0.062	-0.069	-0.065
	(0.010)	(0.010)	(0.011)
Persistence: log tenure	0.299	—	0.443
	(0.009)		(0.016)
Persistence: tenure missing	0.390	0.874	1.986
	(0.638)	(0.510)	(0.349)
Persistence: permanent contract	0.196	0.193	0.306
	(0.012)	(0.012)	(0.014)
Persistence: informal sector	-0.168	-0.172	-0.149
	(0.013)	(0.013)	(0.014)
Error: intercept	-3.966	-4.003	-4.278
	(0.052)	(0.056)	(0.079)
Entry: timegap 4–6 months	—	-0.026	—
		(0.024)	
Entry: timegap 7–12 months	—	-0.096	—
		(0.024)	
Entry: timegap 13–24 months	—	-0.277	—
		(0.024)	
Entry: timegap 25–60 months	—	-0.395	—
		(0.030)	
Entry: timegap over 60 months	—	-0.623	—
		(0.024)	
Persistence: tenure 4–6 months	—	0.044	—
		(0.022)	
Persistence: tenure 7–12 months	—	0.179	—
		(0.022)	
Persistence: tenure 13–24 months	—	0.338	—
		(0.022)	
Persistence: tenure 25–60 months	—	0.498	—
		(0.022)	
Persistence: tenure over 60 months	—	0.679	—
		(0.024)	
Error: age inconsistency	—	—	2.979
			(0.078)
Error: education inconsistency	—	—	1.844
			(0.068)
Error: race inconsistency	—	—	0.787
			(0.123)
Error: gender inconsistency	—	—	2.208
			(0.082)
Error: exactly two inconsistencies	—	—	-0.373
			(0.083)
Error: exactly three inconsistencies	—	—	-1.614
			(0.165)
Error: exactly four inconsistencies	—	—	-2.135
			(0.581)

Appendix Table A5C: Reference piecewise duration hazards

	Entry		Exit	
Duration	Two waves nonemployed	Recent exit	Recent entry	Two waves employed
0–3 months	15.00	43.37	39.13	13.20
	(0.47)	(0.66)	(0.71)	(0.45)
4–6 months	14.42	42.37	37.44	12.28
	(0.44)	(0.92)	(0.90)	(0.40)
7–12 months	12.87	39.62	32.47	9.76
	(0.33)	(0.95)	(0.89)	(0.29)
13–24 months	9.45	32.85	26.96	7.29
	(0.26)	(0.91)	(0.84)	(0.23)
25–60 months	7.62	28.71	21.96	5.32
	(0.34)	(1.06)	(0.76)	(0.17)
Over 60 months	4.85	21.47	16.98	3.63
	(0.14)	(0.75)	(0.68)	(0.13)

Notes. Entries are quarterly probabilities in percent at reference covariate values, with analytical delta-method standard errors in parentheses. Age, age squared, and education are at their standardized means; the reference respondent is male and in the baseline race category. Exit profiles set permanent-contract and informal-sector indicators to zero. Never-worked and missing-duration indicators are also zero. The 0–3 month category is omitted in estimation.

Transition coefficients are on a common probit scale. The unrestricted AR(1) and no-duration AR(2) transition probabilities are re-expressed in their observationally equivalent probit parameterisations; the duration-controlled AR(2) is estimated directly by probit. Initial-state coefficients are logits, and the misclassification coefficient is logit(2π) in all three original models and is reported directly for the staged extension. Parentheses contain analytical sandwich/delta-method standard errors. Duration variables and their missingness indicators vary across origin waves. The information-matrix condition numbers are 14.6 for AR(1), 130.7 for AR(2), and 388,646 for duration-controlled AR(2), 746,558 for the constant-error Set 2 extension, 1,387,680 for the piecewise-duration model, and 7,710,884 for the Table 3 column (1) error model. All three Set 2 models are locally full rank and pass the normalized-score criterion.

Interpretation

- The AR(1) model estimates quarterly entry and exit of 3.20% and 3.13%, with a 2.70% symmetric misclassification probability.
- Duration controls substantially reduce the large AR(2) transition differences but do not eliminate them.
- With Set 2 controls and constant error, entry and exit after two periods in the same state are 4.72% and 4.56%; following a recent reversal they are 31.13% and 32.37%. Observed transition heterogeneity therefore does not explain away the second-lag pattern.
- Piecewise duration hazards improve the log likelihood by 156.4 despite being non-nested with the log-linear model. Entry declines from 15.00% at 0–3 months to 4.85% after 60 months for otherwise reference nonemployed records; exit declines from 13.20% to 3.63% for otherwise reference records employed in both preceding waves. Thus the model finds strong nonlinear duration dependence.
- Nevertheless, the posterior risk-set averages remain 4.73% versus 31.34% for entry and 32.25%

versus 4.59% for exit. Flexible piecewise duration controls therefore do not materially reduce the large AR(2) effects.

- Making misclassification depend on the Table 3 column (1) predictors raises its posterior mean from 0.93% to 1.79%. The corresponding transition rates are 4.32%, 28.98%, 30.18%, and 4.03%, so the pronounced second-lag pattern remains after applying the same observed reliability controls used in Table 3.
- Table 8 shows that the decomposition is specification-sensitive. The Table 4 constant- and reliability-error models attribute 43.0% and 49.4% of apparent transitions to classification, compared with 22.1% in the basic AR(2), 38.5% in the reliability-adjusted AR(2), 31.5% in the common-error FMM, and 27.5% in the type-intercept FMM. The corresponding genuine-reversal shares range from 6.1–6.5% in Table 4 to 18.0–24.3% in the AR(2) and FMM models. The Table 7 full-duration posterior assigns 50.8% to classification, 4.2% to reversal, and 45.0% to persistence, but is not directly status-pair-only because tenure and timegap reports help classify the latent path.
- Surviving AR(2) dependence may reflect unobserved duration, state dependence, latent heterogeneity, or within-quarter transitions; the current estimates do not distinguish these mechanisms.
- The extension is locally identified and has very small final scores, but its rising information condition number calls for sensitivity checks before it is treated as the preferred specification.

To do

- Compare piecewise hazards with spline duration controls and allow the duration profiles to interact with the second-lag state.
- Profile the weakly curved directions in the two Set 2 extensions and check stability with additional long starts and alternative covariate scaling.
- Compare the staged extensions on a common sample with the simpler AR(1), AR(2), and duration-controlled benchmarks.
- With additional waves, test latent heterogeneity, wave-varying transitions, and correlated classification errors as competing explanations.

6. Unobserved heterogeneity: Table 6 and Appendix Table A6

- Re-estimated the two-type finite-mixture model by direct likelihood maximisation with multiple starts and audited local identification.
- Removed the underidentified three-wave FMM table: only the no-error model with free initial conditions is locally identified; symmetric error cannot be separated from two-type transition heterogeneity.
- Developed four-wave alternatives with observed covariates, duration controls, and class-specific misclassification intercepts.
- Replaced the earlier controlled type-specific-error specification with two transparent variants: intercepts only, and intercepts plus the complete Table 3 column (1) predictor set with slopes constrained to be common across types.

Table 6. Comparison of four-wave two-type FMM specifications

	No controls				Controls					
	No error		Common error		Common error		Type intercepts		Type intercepts + predictors	
	Stable	Mobile	Stable	Mobile	Stable	Mobile	Stable	Mobile	Stable	Mobile
Latent-class share (%)	75.65	24.35	92.13	7.87	89.63	10.37	88.68	11.32	80.78	19.22
	(1.00)	(1.00)	(0.37)	(0.37)	(1.26)	(1.26)	(1.11)	(1.11)	(3.22)	(3.22)
Entry rate (%)	0.46	30.46	2.36	42.16	3.20	27.30	3.20	31.05	2.49	20.79
	(0.18)	(0.75)	(0.07)	(1.15)	(0.12)	(2.49)	(0.11)	(2.42)	(0.21)	(2.42)
Exit rate (%)	0.004	33.63	2.21	47.04	3.01	26.87	3.10	31.10	2.22	18.92
	(0.240)	(0.77)	(0.08)	(1.16)	(0.16)	(1.57)	(0.15)	(1.80)	(0.28)	(1.91)
Misclassification rate (%)	0.00	0.00	1.35	1.35	1.74	1.98	1.01	6.28	1.34	5.33
	(fixed)	(fixed)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)	(1.00)	(0.07)	(0.44)
Initial employment rate (%)	47.73	48.17	47.74	47.95	47.38	50.59	47.33	51.74	46.55	52.98
	(0.18)	(0.45)	(0.13)	(0.74)	(0.22)	(1.75)	(0.17)	(1.27)	(0.32)	(0.97)
Log-likelihood	-451,874.52		-450,805.93		-428,777.56		-439,546.18		-427,570.80	
N	308,240		308,240		308,240		308,240		308,240	

All estimates except the log likelihood and observation count are percentages; analytical survey-weighted sandwich standard errors, transformed by the delta method, are in parentheses. Entry and exit rates are posterior risk-set weighted in the controlled models. Posterior mean misclassification averages each respondent-wave's fitted posterior mismatch probability using survey weights and posterior class membership. Controls comprise observed covariates and duration controls. All controlled models use common transition slopes and class-specific transition intercepts. The fourth model has class-specific misclassification intercepts and no misclassification covariates. The fifth adds the exact Table 3 column (1) predictors—age, education, race, and gender inconsistencies and indicators for exactly two, three, or four inconsistencies—with slopes common across classes. In the no-error model, all misclassification probabilities are fixed at zero. The no-error model has full probability-Jacobian rank, but weak curvature around its near-zero stable-class exit estimate warrants caution. Misclassification coefficients and optimisation diagnostics are reported in Appendix Table A6.

Appendix Table A6. Misclassification equations and diagnostics

Quantity	Type intercepts	Type intercepts + predictors
Stable-type intercept	-3.880 (0.037)	-4.815 (0.091)
Mobile-type intercept	-1.940 (0.183)	-2.834 (0.120)
Age inconsistent	–	3.189 (0.063)
Education inconsistent	–	1.896 (0.055)
Race inconsistent	–	0.745 (0.125)
Gender inconsistent	–	2.343 (0.072)
Exactly two inconsistencies	–	-0.220 (0.085)
Exactly three inconsistencies	–	-1.474 (0.157)
Exactly four inconsistencies	–	-1.313 (1.720)
Likelihood cells	213,784	233,944
Information-matrix rank	30/30	37/37
Minimum information eigenvalue	0.0000174	0.00000217
Information condition number	21,309	163,747
Maximum weighted score	0.00000484	0.00000669

Coefficients enter the index of the symmetric misclassification probability, $0.5\,\mathrm{logit}^{-1}(z\delta)$. Standard errors in parentheses are analytical survey-weighted sandwich standard errors. Both specifications use the same 308,240-person controlled sample. The different cell counts reflect the additional error-equation predictors, not a different estimation sample.

Interpretation

- The three-wave symmetric-error FMM is underidentified and should not be reported structurally; four waves are required to separate the two mechanisms.
- Imposing no misclassification pushes the stable-class exit estimate to essentially zero and attributes 30–34% quarterly transition rates to the mobile class. Allowing error materially improves fit and avoids this extreme decomposition.
- Four waves support a large stable/mobile distinction after controlling for observed characteristics and duration.
- With type-specific error intercepts alone, the posterior mean error rate is 1.01% for the stable class and 6.28% for the mobile class. Adding the Table 3 predictors changes these to 1.34% and 5.33%, while increasing the estimated mobile share from 11.32% to 19.22%.
- The richer error equation substantially improves fit and produces less extreme mobile-class entry and exit rates (20.79% and 18.92%) than the intercept-only model (31.05% and 31.10%). Thus some heterogeneity otherwise assigned to class membership is associated with observed report inconsistencies and matching-quality signals.
- The type-specific probabilities describe joint latent classifications; they do not show that mobility causes reporting or matching error.
- The four-wave results are best presented as a replacement or appendix extension rather than as validation of the underidentified three-wave results.

To do

- Complete profile-likelihood, multistart, and simulation checks for the four-wave model's identification and class separation.
- Test covariate-dependent class membership and, with a longer panel, whether latent types can be combined with AR(2) state dependence.

7. Endogenous duration dependence: Table 7

- Modelled transition hazards as functions of endogenous tenure and time since work, separately from the age/education-consistency analysis in Table 3.
- Repaired the tenure-contamination likelihood and optimisation diagnostics.
- Retained exact consistency between clean duration reports and latent employment histories.
- Integrated unavailable wrong-state duration clocks instead of imputing them.
- Replaced constant and power-law transition hazards with tail-safe piecewise hazards.
- Added contamination in reported time since work and tested local and joint alternatives.
- Split tenure contamination into bounded local month errors and gross marginal draws, with a boundary-aware likelihood check.

Table 7: Tenure-contamination misclassification and transition rates

	Constant, exact	Power law, exact	Piecewise, exact	Independent error	Local error	Joint per-wave error	Local + gross tenure
Entry rate	4.08	1.96	3.64	3.13	2.91	3.15	3.15
	(0.01)	(0.01)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)
Exit rate	4.26	4.51	4.62	4.42	4.43	4.42	4.42
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Misclassification rate	10.21	10.53	9.93	2.33	3.02	2.32	2.32
	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.02)
Initial employment rate	54.98	55.92	55.02	48.30	48.75	48.30	48.30
	(0.08)	(0.08)	(0.09)	(0.08)	(0.08)	(0.08)	(0.08)
Tenure contamination rate	24.79	25.09	24.98	25.22	25.22	25.21	25.21
	(0.07)	(0.07)	(0.07)	(0.06)	(0.06)	(0.06)	(0.06)
Timegap contamination rate	0.00	0.00	0.00	28.04	39.20	16.62	16.62
	fixed	fixed	fixed	(0.11)	(0.15)	(0.07)	(0.07)
Log likelihood	-2,510,430.3	-2,475,676.8	-2,462,197.4	-2,330,075.3	-2,343,051.4	-2,320,294.9	-2,320,294.9
N	402,923	402,923	402,923	402,923	402,923	402,923	402,923
Transition-hazard form	Constant	Power law	Piecewise	Piecewise	Piecewise	Piecewise	Piecewise
Tenure-error model	Gross marginal	Gross marginal	Gross marginal	Gross marginal	Gross marginal	Gross marginal	Local + gross marginal
Timegap-error model	None (exact)	None (exact)	None (exact)	Independent marginal	Local category	Joint per-wave marginal	Joint per-wave marginal

Notes. Rates are in percent and follow the common entry, exit, misclassification, and initial-employment order used in the other main tables; model-specific duration-error rates follow. Entry and exit are posterior risk-set and survey-weighted quarterly probabilities. Parentheses contain numerical observed-information/delta-method standard errors from the full covariance matrix. Gross tenure errors are independent marginal draws. The final model also permits symmetric local errors of one to six months with fixed exponentially declining weights; its local probability is estimated at the zero boundary, so common standard errors equal those in the preceding nested column. 'Exact' means that no separate timegap-contamination process is allowed.

For the joint model, 16.62% is the per-report probability. It implies a 30.48% probability that at least one report in a two-report pair is contaminated. The local-plus-gross extension estimates zero local tenure contamination and 25.21% gross contamination, so its other point estimates equal the preferred joint per-wave model.

Appendix Table A7, Panel A. Exact and alternative duration profiles

Duration interval	Exact exit hazard	Exact exit (%)	Independent-error exit (%)	Joint-model exit (%)
0–3 months	0.3420	8.35	7.79	7.79
3–12 months	0.3558	8.51	7.90	7.90
1–3 years	0.1757	4.30	4.23	4.23
3–5 years	0.1522	3.73	3.67	3.67
5+ years	0.1353	3.33	3.27	3.27

Duration interval	Exact entry hazard	Exact entry (%)	Independent-error entry (%)	Joint-model entry (%)
0–3 months	0.5060	9.15	7.66	7.88
3–12 months	0.2620	6.34	5.00	5.07
1–3 years	0.1098	2.71	2.27	2.27
3–5 years	0.1177	2.90	2.57	2.57
5+ years	0.1425	3.50	2.82	2.83

Appendix Table A7, Panel B. Auxiliary model comparisons

Quantity	Constant, exact	Power law, exact	Piecewise, exact
Parameters	5	7	13
AIC	5,020,870.7	4,951,367.6	4,924,420.8
Mean employment duration (months)	68.9	68.1	67.0
Mean nonemployment duration (months)	72.0	1,257.2	76.7
Comparable two-report timegap contamination (%)	0	0	0
Analytical SE	fixed	fixed	fixed

Quantity	Independent error	Local error	Joint per-wave error
Parameters	14	14	14
AIC	4,660,178.6	4,686,130.8	4,640,617.9
Mean employment duration (months)	69.6	69.6	69.6
Mean nonemployment duration (months)	95.6	102.2	94.8
Comparable two-report timegap contamination (%)	28.04	39.20	30.48
Analytical SE	0.11	0.15	0.12

Quantity	Local + gross tenure
Parameters	15 (14 interior)
AIC	4,640,619.9
Total tenure contamination (%)	25.21
Local tenure contamination (%)	0.00 (boundary)
Gross tenure contamination (%)	25.21
Comparable two-report timegap contamination (%)	30.48

Appendix Table A7, Panel C. Effect of integrating unavailable wrong-state clocks

Quantity	Imputed clock	Integrated clock
Entry rate (%)	2.08	1.96
Exit rate (%)	4.91	4.51
Misclassification rate (%)	10.35	10.53
Tenure contamination rate (%)	24.81	25.09
Exit duration slope	-0.4413	-0.3742
Entry duration slope	-0.9148	-0.8782
Mean nonemployment duration (months)	3,800	1,257

Appendix Table A7, Panel D. Timegap-contamination mechanism decomposition

Interview interval	Clock-feasible (%)	Reset-compatible contradiction (%)	Other contradiction (%)	Posterior contamination (%)
Wave 1 to wave 2	84.42	2.82	12.76	28.20
Wave 2 to wave 3	84.95	2.98	12.07	27.87

In the power-law model, exit falls from 7.42% at zero tenure to 3.22% at ten years, while entry falls from 8.51% to 1.18%. Integrating rather than imputing wrong-state clocks affects 15.5% of the posterior employment risk set and 4.5% of the nonemployment risk set. In the piecewise exact model, median employment and nonemployment durations are 35.3 and 51.4 months. For the independent-error model, clock-impossible shares are 15.6% and 15.0%, while average contaminated-branch probabilities among feasible reports are 14.96% and 15.09%. Roughly 10% of fitted contamination mass is reset-compatible, 43–45% comes from contradictions that a hidden reset cannot explain, and 45–46% is allocated among clock-feasible reports. Joint-model exit risks differ from the independent-error risks by less than 0.03 percentage points in every interval.

The independent-error information matrix has rank 14/14, condition number 45, and a sharply peaked likelihood around 28.04%; the boundary likelihood-ratio diagnostic is 264,244. Refined maximum scaled gradients are 2.0×10^{-4} for the exact piecewise fit, 2.1×10^{-6} for the local-error fit, and 2.5×10^{-6} for the joint model. These are numerical diagnostics, not substitutes for the standard errors reported in Table 7. For the local-plus-gross tenure extension, the likelihood falls by 4.0 even when only 0.1% of total tenure contamination is reassigned to the local branch. All 18 dispersed positive-share candidates fit worse than the boundary.

Interpretation

- Piecewise hazards fit much better than constant or power-law hazards and improve the weighted-likelihood AIC by about 26,947 relative to the power law, while avoiding its implausibly long nonemployment tail.
- Exit falls after the first year of tenure. Entry falls sharply with time since work and then levels off rather than approaching zero.
- Allowing timegap contamination reduces estimated status error from about 10% to 2–3%, while tenure contamination remains close to 25%.
- The joint per-wave model fits best and leaves transition rates and duration profiles almost unchanged relative to the simpler contamination model.
- The proposed one-to-six-month local tenure-error component receives zero probability. The data prefer to retain the full 25.21% tenure-contamination rate as gross marginal error; adding the unused component raises AIC by two and leaves all substantive estimates unchanged.
- Only about 3 percentage points of timegap contradictions can be explained by an unobserved within-quarter employment spell; hidden resets are not the main source of the contamination estimate.
- The results identify inconsistency among status, tenure, timegap, and linked histories. They do not determine whether the underlying cause is respondent error, matching error, or another data problem.

To do

- Add bootstrap or profile uncertainty and posterior-predictive checks for the joint status and duration distributions.
- Test alternative contamination densities, rounding and heaping rules, and tail cut-points while retaining transition-duration consistency.
- Replace the symmetric local month kernel with explicit six- and twelve-month heaping operators before concluding that all non-exact tenure reports are genuinely uninformative about the latent clock.

8. Apparent-transition implications: Table 8

Table 8: Implications for apparent two-wave transitions

Model	Classification only	Genuine reversal	Genuine persistence
All apparent transitions			
Table 1 (1): raw/no error	0.0	8.4	91.6
Table 1 (2): symmetric error	67.0	1.0	32.1
Table 4 (4): Set 2, constant error	43.0	6.5	50.5
Table 4 (5): Set 2, reliability predictors	49.4	6.1	44.5
Table 5 (2): AR(2)	22.1	24.3	53.6
Table 5 (6): AR(2), reliability predictors	38.5	18.0	43.5
Table 6 (3/4): FMM, common error	31.5	19.6	48.8
Table 6 (7/8): FMM, type intercepts	27.5	18.7	53.9
Table 7 (4): independent duration error*	50.8	4.2	45.0
Table 9 (1): duration AR(1)*	56.4	6.0	37.6
Table 9 (2): duration AR(2)*	50.8	12.1	37.1
Apparent entries			
Table 1 (1): raw/no error	0.0	8.5	91.5
Table 1 (2): symmetric error	66.0	1.0	33.0
Table 4 (4): Set 2, constant error	43.8	6.7	49.5
Table 4 (5): Set 2, reliability predictors	47.7	6.4	45.9
Table 5 (2): AR(2)	21.8	24.7	53.5
Table 5 (6): AR(2), reliability predictors	36.3	19.0	44.7
Table 6 (3/4): FMM, common error	31.0	20.2	48.7
Table 6 (7/8): FMM, type intercepts	27.4	19.2	53.4
Table 7 (4): independent duration error*	51.9	3.2	44.9
Table 9 (1): duration AR(1)*	61.7	6.5	31.8
Table 9 (2): duration AR(2)*	55.7	14.0	30.3
Apparent exits			
Table 1 (1): raw/no error	0.0	8.3	91.7
Table 1 (2): symmetric error	67.9	1.0	31.1
Table 4 (4): Set 2, constant error	42.2	6.3	51.4
Table 4 (5): Set 2, reliability predictors	51.1	5.7	43.2
Table 5 (2): AR(2)	22.5	23.8	53.7
Table 5 (6): AR(2), reliability predictors	40.8	17.0	42.2
Table 6 (3/4): FMM, common error	32.1	19.0	48.9
Table 6 (7/8): FMM, type intercepts	27.5	18.1	54.4
Table 7 (4): independent duration error*	49.7	5.2	45.2
Table 9 (1): duration AR(1)*	50.9	5.5	43.6
Table 9 (2): duration AR(2)*	45.7	10.1	44.2

Notes. Entries are survey-weighted percentages. Unstarred rows condition only on the two status reports forming an apparent transition and integrate out the subsequent report. Classification only means reported status changes but latent status does not. Genuine reversal means latent status changes and returns next period; genuine persistence means it remains in the new state. These outcomes are mutually exclusive and exhaustive; a latent change opposite to the reported direction is still a genuine change. Paired Table 6 references are the two types of one FMM, integrated over membership. *Table 7 and Table 9 use full-record posteriors including duration evidence, not status-pair-only conditioning. Table 9 pools apparent transitions 1–2 and 2–3 with their follow-up waves, using all four reports. Its rates are conditional on apparent switching, not the population status-error probability. Models use their own samples. Table 9 uses the current unrevised onset and employer-return assumptions; its uncertainty is pending.

This comparison is deliberately placed after the model-specific tables because it synthesizes their implications for interpreting apparent transitions.

The new four-wave duration models in the separate candidate Table 9 document use the same Panel A sample. Update their rows without re-estimation by running:

```
Rscript scripts/update_table8_duration_models.R
```

Unlike the earlier audit's "not confirmed in the reported direction" measure, Table 8 counts any latent status change as genuine, even if its direction is opposite to the reported change. The long summary includes this same table.

9. No-error duration counterfactual (Appendix Table A8)

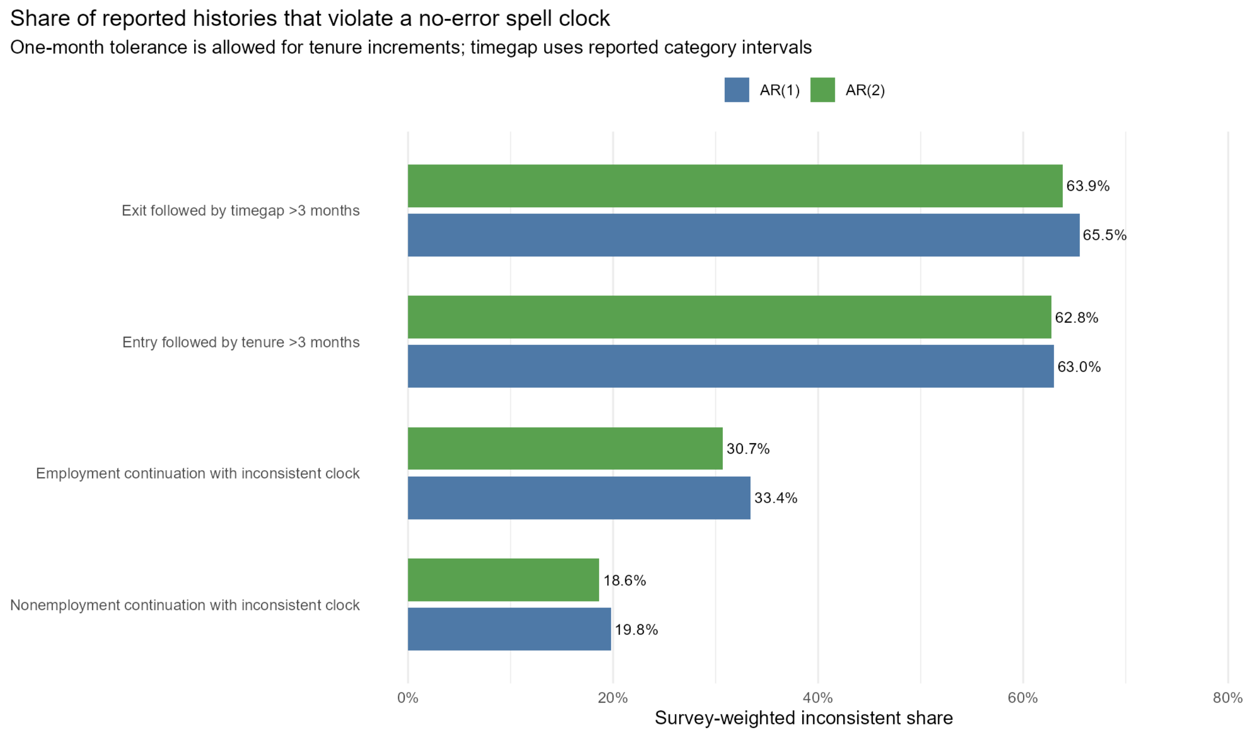
- Estimated flexible AR(1) and AR(2) transition models treating reported status changes as error-free.
- Simulated the duration clocks implied by those transitions.
- Compared simulated and observed tenure and timegap distributions, with transition-conditional graphs as the primary diagnostic.

Appendix Table A8: No-error duration-clock contradictions

	AR(1)	AR(2)
Apparent entry followed by tenure over three months	63.0	62.8
	(0.30)	(0.35)
Apparent exit followed by timegap over three months	65.5	63.9
	(0.29)	(0.35)
Employment continuation with inconsistent tenure	33.4	30.7
	(0.11)	(0.13)
Nonemployment continuation with unreachable timegap	19.8	18.6
	(0.09)	(0.10)

Notes. Entries are survey-weighted shares in percent. Parentheses contain person-clustered linearized standard errors. Each no-error model allows flexible duration dependence in observed transitions; destination-wave durations are held out and used for these clock-consistency diagnostics.

Graph accompanying Panel A. Clock contradictions under AR(1) and AR(2)

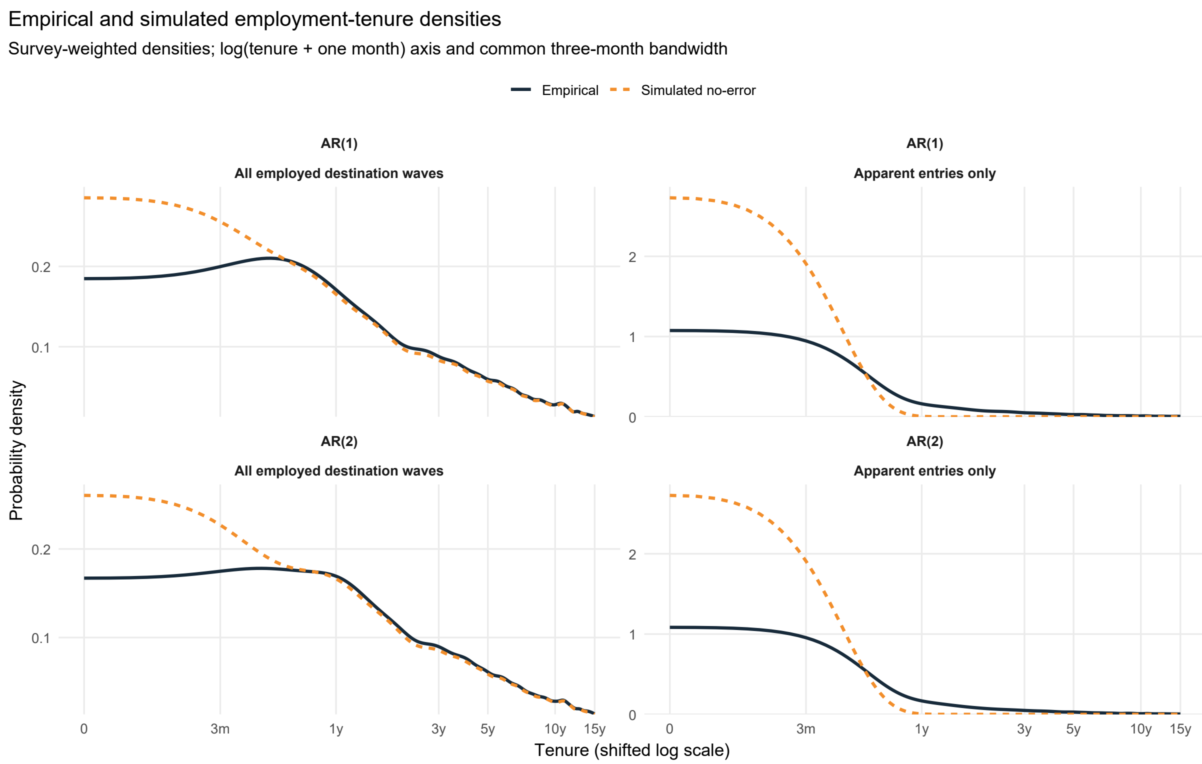


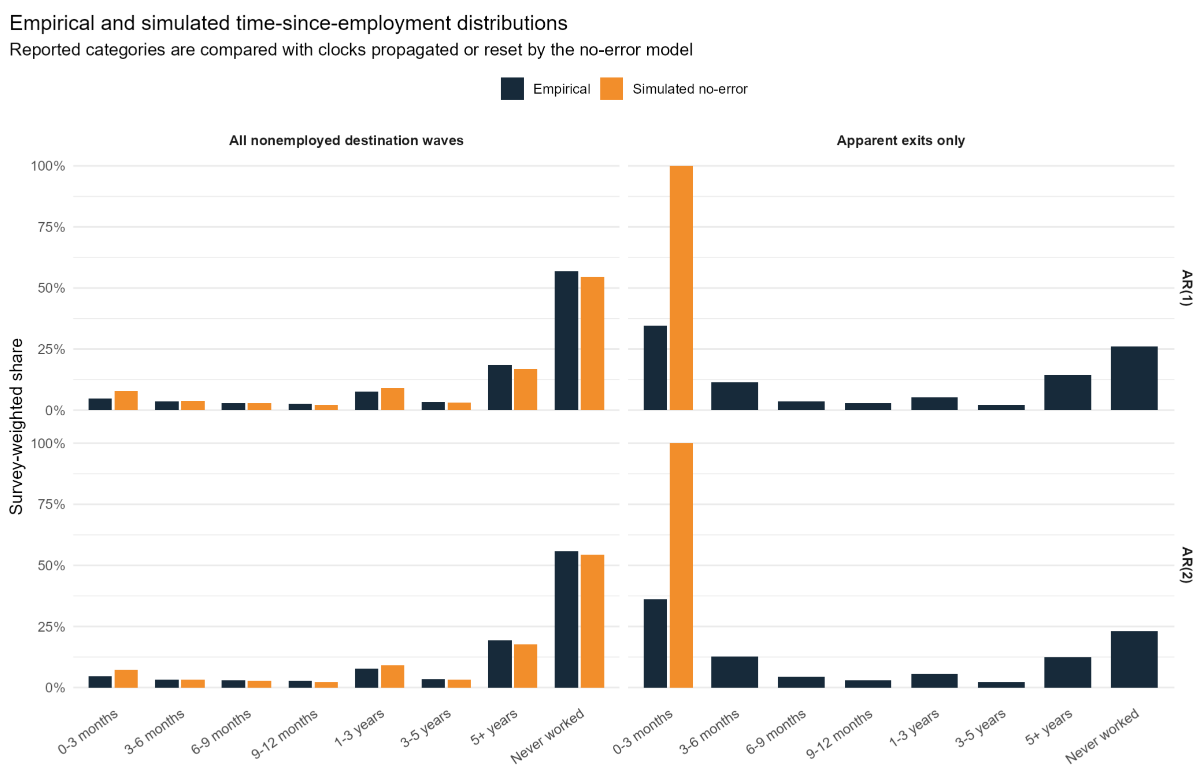
Appendix Table A8, Panel B. Observed transition benchmarks

Rate	AR(1) (%)	AR(2) (%)
Entry rate	8.23	7.78
Exit rate	8.47	7.83
Admissible tenure after apparent entry	37.0	37.2

The flexible transition equations reproduce these observed entry and exit rates. Among apparent exits in the AR(1) analysis, 26.0% report that they have never worked, reinforcing the contradiction between status changes and the reported duration history.

Graphs accompanying Panel B. Empirical and simulated duration distributions


Empirical and simulated employment-tenure densities
Survey-weighted densities; log(tenure + one month) axis and common three-month bandwidth
Empirical
Simulated no-error
AR(1)
All employed destination waves
AR(1)
Apparent entries only
AR(2)
All employed destination waves
AR(2)
Apparent entries only
Probability density
0.2
0.1
2
1
0
0
3m
1y
3y
5y
10y
15y
Tenure (shifted log scale)


Empirical and simulated time-since-employment distributions
Reported categories are compared with clocks propagated or reset by the no-error model
Empirical
Simulated no-error
All nonemployed destination waves
Apparent exits only
AR(1)
AR(2)
Survey-weighted share
100%
75%
50%
25%
0%
0-3 months
3-6 months
6-9 months
9-12 months
1-3 years
3-5 years
5+ years
Never worked

Interpretation

- Flexible no-error transition models reproduce average observed changes but cannot reconcile them with destination-wave duration reports.
- Most apparent entries and exits imply duration clocks that are impossible under a genuine transition during the interview interval.
- Marginal duration distributions obscure the problem because most respondents do not change reported state; transition-conditional comparisons are more persuasive.
- The diagnostic motivates jointly modelling status and duration consistency without claiming which reported variable or linked record is incorrect.

To do

- Add simulation uncertainty bands to the empirical-versus-implied duration graphs and report transition-conditional comparisons prominently.
- Repeat the counterfactual under alternative matching panels and initial-clock assumptions.

Reproducibility and remaining inference

- A result-by-result audit against `branch_changes_summary.Rmd` maps the baseline estimates to Table 1/A1, matching-rule results to Table 2, age/education-consistency results to Table 3/A2, covariate estimates and hazard distributions to Table 4/A4, AR(2) results to Table 5/A5, four-wave FMM results to Table 6/A6, tenure-contamination results to Table 7/A7, and the no-error duration benchmarks to A8. No numerical estimate reported in the long summary is intentionally omitted here; only detailed implementation narrative is abbreviated.
- Tables 1, 3, and 4 and Appendix Tables A1–A4 are rebuilt by build_tables.R.
- Table 2 is rebuilt by the matching-sensitivity script; the four-wave, tenure, and diagnostic scripts save the inputs for Tables 5–7 and Appendix Tables A5–A8 outside version control.
- Analytical standard errors are available for Tables 1–7 and their associated model estimates. Table 7 uses the full numerical observed-information matrix and delta method; Appendix Table A8 uses person-clustered linearization for its weighted descriptive shares.
- Table 7's numerical observed-information errors should still be checked with a person-level bootstrap or profile likelihood before final submission.
- A person-level bootstrap remains the most useful common inference robustness check across the paper.